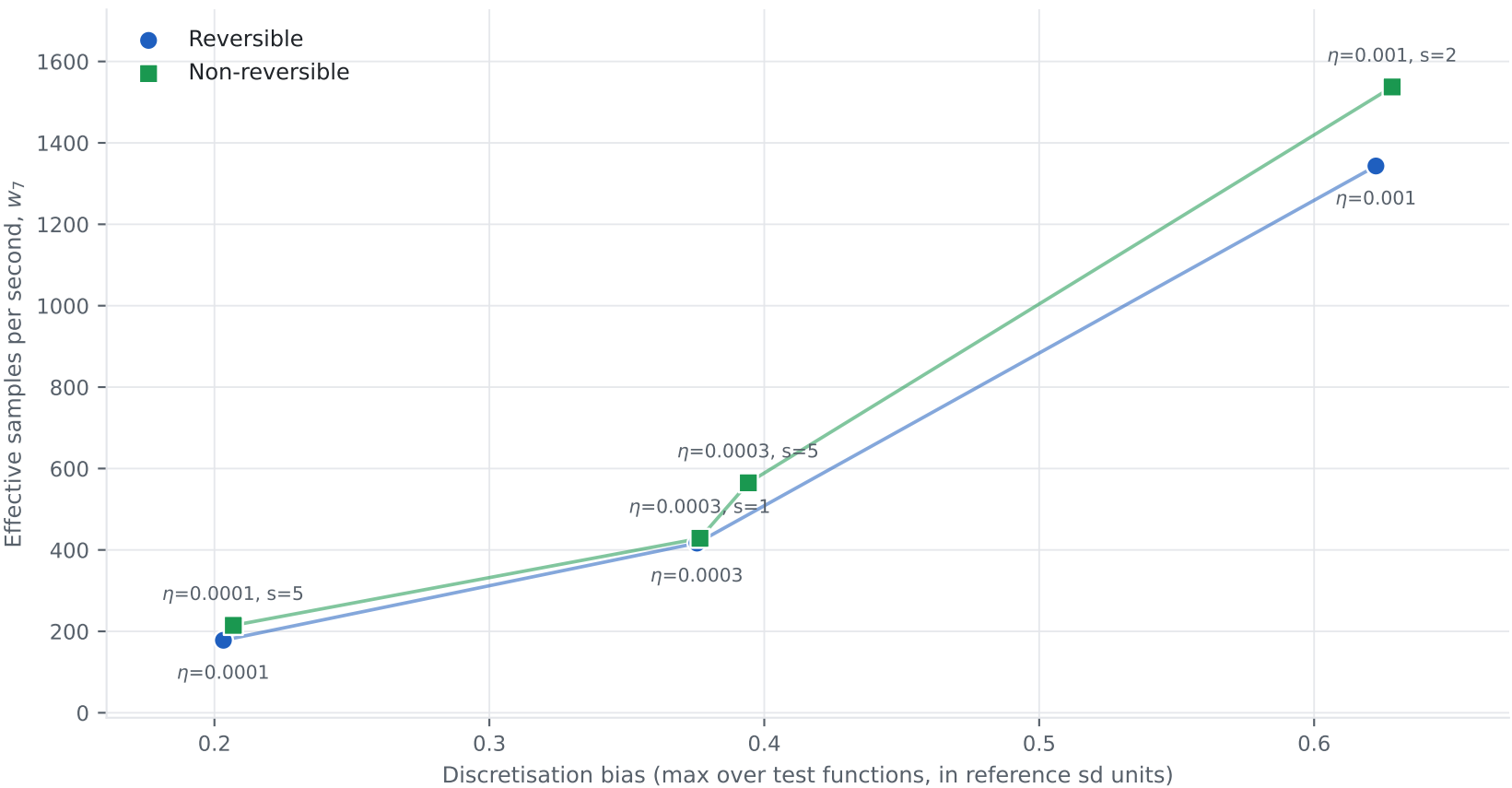


At matched bias the non-reversible arm is strictly more efficient



Each point is one (arm, eta, s) configuration; both arms are free to choose their own step size, which is what makes this a best-versus-best comparison. Bias is measured against an $\eta \rightarrow 0$ extrapolation of the reversible ensemble mean, not against a single small-eta run. Reading vertically at any bias level the green curve sits above the blue one. Comparing points at DIFFERENT bias is not a fair comparison: a larger step buys efficiency by paying in bias, and both arms can do that. Seed 5400, R = 96, 16000 iterations.